

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 01: EXPLORATION: ENTERING THE WORLD OF SECONDARY SCIENCE
TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-01

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
5. **Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): The International System of Units (SI) is adopted universally across global scientific communities.

Reason (R): The SI system is built upon coherent, standardized base-10 decimal units that eliminate arbitrary regional conversion errors.

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): A simple pendulum of mass 50 g and another of mass 100 g having identical length exhibit the exact same time period of oscillation at a given place.

Reason (R): The time period of a simple pendulum is strictly independent of the mass and material of the oscillating bob.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): Relative density of a solid substance has no physical unit.

Reason (R): Relative density is defined as the ratio of the density of the substance to the density of water at 4°C, causing all units to cancel out.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): A measurement of 5.00 cm is more precise than a measurement of 5.0 cm.

Reason (R): Precision is determined by the least count of the measuring instrument; 5.00 cm indicates measurement up to two decimal places.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): The period of oscillation of a simple pendulum increases when taken to the top of Mount Everest.

Reason (R): The acceleration due to gravity g decreases with increasing altitude above the Earth's surface.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): Parallax error occurs when an observer views the pointer or meniscus of a measuring instrument at an oblique angle.

Reason (R): To obtain an accurate reading, the observer's line of sight must always be positioned perpendicularly to the instrument scale.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.7

Assertion (A): The density of an iron nail is greater than the density of a massive wooden log.

Reason (R): Iron atoms possess greater atomic mass and are packed much more densely in a crystalline lattice than organic wood fibers.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.8

Assertion (A): A piece of cork floats on water, whereas an iron nail of the same volume sinks immediately.

Reason (R): Water exerts an upward buoyant force only on light objects like cork and exerts zero buoyant force on heavy objects like iron.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.9

Assertion (A): In scientific laboratory investigations, multiple trials of an experiment are recorded and averaged.

Reason (R): Averaging repeated measurements minimizes random observational errors and enhances experimental reliability.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.10

Assertion (A): When measuring the volume of water in a graduated cylinder, the reading should be taken at the upper convex meniscus.

Reason (R): Water wets the glass surface, forming a concave meniscus due to strong adhesive forces between water and glass molecules.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)

[8 MARKS]

Q.11

Assertion (A): A graph plotted between T^2 (square of time period) and L (length) for a simple pendulum is a straight line passing through the origin.

Reason (R): The square of the time period of a simple pendulum is directly proportional to its effective length ($T^2 \propto L$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): A vernier caliper can measure internal diameter, external diameter, and depth of a hollow cylinder.

Reason (R): A vernier caliper is equipped with upper jaws, lower jaws, and a metallic depth probe strip.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): If a pendulum clock is transported from the North Pole to the Equator, it runs slower and loses time.

Reason (R): The value of g is slightly smaller at the Equator than at the Poles due to the Earth's equatorial bulge and rotational centrifugal effect.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): Zero error in a screw gauge is a systematic error that can be mathematically corrected by addition or subtraction.

Reason (R): Systematic errors consistently skew measurements in a single direction and arise from faulty calibration or zero offset.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): A wire gauze with an asbestos or ceramic center is placed between a Bunsen burner flame and a glass beaker.

Reason (R): Direct exposure to concentrated flame causes localized uneven expansion of glass, leading to thermal fracture.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): A body of mass 10 kg has a weight of 98 N on Earth and a weight of 0 N in deep interstellar space.

Reason (R): Mass is an invariant intrinsic property of matter, while weight is a gravitational force dependent on local acceleration due to gravity.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): Water can never be used to extinguish a burning oil or petrol fire in the laboratory.

Reason (R): Water reacts violently with petrol to produce toxic chlorine gas.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): A block of wood floats with 60% of its volume submerged in water. Its density is 0.6 g/cm^3 .

Reason (R): According to the principle of floatation, a floating body displaces a volume of liquid whose weight equals the total weight of the body.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)

[7 MARKS]

Q.19

Assertion (A): The amplitude of oscillation of a simple pendulum in a school lab must be kept small (under 10°).

Reason (R): The mathematical formula $T = 2\pi \sqrt{L/g}$ is strictly derived using the small-angle approximation $\sin \theta \approx \theta$.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): While heating a test tube containing chemicals, its mouth should never be directed towards oneself or lab partners.

Reason (R): Boiling liquids in test tubes can superheat and violently eject hot reactive chemicals due to sudden vapor bubble formation (bumping).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): If the length of a simple pendulum is increased by four times, its time period doubles.

Reason (R): The time period of a simple pendulum is inversely proportional to the square root of its length.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): A hydrometer sinks deeper in kerosene than in pure water.

Reason (R): Kerosene has a lower density than water, requiring a larger volume of kerosene to be displaced to balance the hydrometer's weight.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): The least count of an instrument can be decreased indefinitely without any practical physical limit.

Reason (R): Decreasing least count always increases the precision of any measurement.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): During an experiment to determine the boiling point of water, pumice stones are added to the boiling flask.

Reason (R): Pumice stones introduce nucleation sites, promoting steady, smooth boiling and preventing explosive superheating.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): In a controlled scientific experiment, only one independent variable should be altered at a time while holding all other variables constant.

Reason (R): If multiple variables are changed simultaneously, it becomes scientifically impossible to determine which specific factor caused the observed effect.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(A)	Assertion is true. Reason is true and correctly explains why SI units are universally standard across scientific disciplines.
Q.2	(A)	Assertion is true. Reason is true and directly explains the formula $T = 2\pi \sqrt{L/g}$, which contains no mass term.
Q.3	(A)	Assertion is true. Reason is true and correctly explains why a ratio of two identical physical quantities is dimensionless.
Q.4	(A)	Assertion is true. Reason is true and correctly explains that higher decimal resolution reflects finer instrument least count.
Q.5	(A)	Assertion is true. Reason is true and correctly explains the formula $T = 2\pi \sqrt{L/g}$; as g decreases at high altitudes, T increases.
Q.6	(B)	Both statements are scientifically true, but Reason states the prevention procedure rather than explaining the optical cause of parallax displacement.
Q.7	(A)	Assertion is true. Reason is true and provides the microscopic structural explanation for the higher macroscopic density of iron.
Q.8	(C)	Assertion is true. Reason is false: Water exerts buoyant force on all submerged objects; the nail sinks because its weight exceeds the buoyant force.
Q.9	(A)	Assertion is true. Reason is true and correctly explains the statistical purpose of taking repeated readings.
Q.10	(D)	Assertion is false (water readings must be taken at the lowest point of the concave meniscus). Reason is true (water forms a concave meniscus due to adhesive forces).
Q.11	(A)	Assertion is true. Reason is true and correctly explains the linear relationship $T^2 = (4\pi^2/g)L$ represented on the graph.
Q.12	(A)	Assertion is true. Reason is true and directly explains the mechanical construction enabling multi-dimensional measurements.
Q.13	(A)	Assertion is true. Reason is true and explains why T increases at the equator, causing the pendulum to oscillate fewer times per day (losing time).
Q.14	(A)	Assertion is true. Reason is true and correctly defines systematic errors and justifies algebraic correction.
Q.15	(A)	Assertion is true. Reason is true and explains how the wire gauze conducts and disperses heat uniformly to prevent thermal shock.
Q.16	(A)	Assertion is true. Reason is true and clearly explains the fundamental distinction between mass and weight in different gravitational fields.
Q.17	(C)	Assertion is true. Reason is false: Water does not react chemically to make chlorine gas; rather, petrol is lighter than water and floats on top, spreading the fire.
Q.18	(A)	Assertion is true ($V_{\text{sub}}/V = \rho_{\text{body}}/\rho_{\text{liquid}} = 0.60$). Reason is true and correctly provides the governing law of floatation.
Q.19	(A)	Assertion is true. Reason is true and explains the mathematical requirement for simple harmonic motion.
Q.20	(A)	Assertion is true. Reason is true and explains the physical hazard of sudden boiling/bumping in narrow glassware.
Q.21	(C)	Assertion is true ($T \propto \sqrt{L}$; $\sqrt{4} = 2$). Reason is false: T is directly proportional to $\sqrt{L}$, not inversely proportional.
Q.22	(A)	Assertion is true. Reason is true and correctly explains how hydrometers indicate lower liquid density by greater depth of submersion.

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.23	(D)	Assertion is false (material imperfections, thermal expansion, and quantum/optical diffraction impose strict physical limits). Reason is true (smaller least count yields higher precision).
Q.24	(A)	Assertion is true. Reason is true and explains the functional mechanism of anti-bumping boiling granules.
Q.25	(A)	Assertion is true. Reason is true and explains the fundamental criterion of scientific fair testing.